

DIGITAL MEDIA CONTENT PUBLISHING MATRIX

VidHub – A DSA-Based Video Streaming Platform

ITM Skills University

Department of Computer Science Engineering
Subject: Data Structures and Algorithms (DSA)

Student Details

Name: Ayush Mishra
PRN: 150096725046
Course: B.Tech CSE
Cohort: Jeff Bezos

3.3 Project Title

Digital Media Content Publishing Matrix – VidHub

3.4 Problem Statement

VidHub is a DSA-based video streaming platform that organizes video catalogs, tracks watch history, validates licenses, ranks content, manages recommendations, models sharing networks, optimizes delivery paths, and supports video segmentation.

3.5 Objectives

- Maintain an organized video catalog.
- Track watch history.
- Manage recommendation queues.
- Verify copyright licenses.
- Rank videos by engagement.
- Model sharing networks.
- Optimize delivery paths.
- Segment videos for streaming.

3.6 Design / Architecture

Video Catalog → BST
Watch History → Stack
Recommendation Queue → Queue
License Registry → Hash Map
Ranking System → Max Heap
Sharing Network → Graph
Delivery Optimizer → Dijkstra Algorithm
Video Segment Manager → Deque

3.7 Algorithm Description

BST for catalog management, Stack for history, Queue for recommendations, Hash Map for license validation, Heap for ranking, Graph + Dijkstra for network optimization, and Deque for segment management.

3.8 Complexity Analysis

Search (BST): $O(\log n)$
Stack Push: $O(1)$
Queue Enqueue: $O(1)$
Hash Lookup: $O(1)$
Heap Insert: $O(\log n)$
Dijkstra: $O(E \log V)$

3.9 Screenshots of Execution

Include the screenshots collected for: Login Screen, Creator Dashboard, License Validation, Video Catalog, Playback Screen, Watch History, and Recommendation Queue.

3.10 Results and Findings

The system successfully demonstrates practical applications of DSA concepts used in modern streaming platforms. Efficient retrieval, ranking, recommendation, and delivery optimization were achieved.

3.11 Conclusion

VidHub successfully integrates multiple data structures and algorithms into a unified video streaming simulation, showcasing their real-world relevance.

3.12 GitHub Repository Link

<https://github.com/Ayusshh020/VidHub>